

Quiz 11

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (10 points) The prices of a security follow a geometric Brownian motion with drift parameter $\mu = 0.12$ and volatility parameter $\sigma = 0.24$. The security's present price is 50. There is a call option having six months until its expiration time and with a strike price of 52.

(a) What is the probability of that the call option will be exercised?

$$\begin{aligned}
 S &= 50, K = 52, t = \frac{6}{12} = \frac{1}{2} \\
 P(S(\frac{1}{2}) > 52) &= P\left(\frac{S(\frac{1}{2})}{S(0)} > \frac{52}{50}\right) = P(X > \log(1.04)) \\
 &= P\left(\frac{X - 0.06}{0.1697} > \frac{\log(1.04) - 0.06}{0.1697}\right) \quad \begin{array}{l} X \sim N(\mu t, \sigma^2 t) \\ \mu t = 0.12 \times \frac{1}{2} = 0.06 \\ \sigma \sqrt{t} = 0.24 \times \sqrt{\frac{1}{2}} \approx 0.1697 \end{array} \\
 &= P(Z > -0.1224) \\
 &= P(Z < 0.1224) \\
 &= \Phi(0.1224) \approx 0.5478
 \end{aligned}$$

(b) Assume that the nominal interest rate is 6%. Find the no-arbitrage cost of the call option by using the Black-Scholes formula.

$$\begin{aligned}
 r &= 0.06, t = \frac{1}{2}, S = 50, K = 52, \sigma = 0.24 \\
 d_1 &= \frac{rt + \sigma^2 t/2 - \log(K/S)}{\sigma \sqrt{t}} = \frac{0.06 \times \frac{1}{2} + (0.24)^2 \times \frac{1}{2} \times \frac{1}{2} - \log(\frac{52}{50})}{0.24 \times \sqrt{\frac{1}{2}}} \\
 &\approx 0.03052 \\
 \Rightarrow C &= S \Phi(d_1) - K e^{-rt} \Phi(d_1 - \sigma \sqrt{t}) \\
 &= 50 \Phi(0.03052) - 52 e^{-0.06/2} \Phi(0.03052 - 0.24 \sqrt{1/2}) \\
 &= 50 \Phi(0.03052) - 50.462 \Phi(-0.1392) \\
 &\approx 25.6 - 50.462 \times (1 - 0.5557) \\
 &= 3.18
 \end{aligned}$$